

System Identification of the Sound xR Image Renderer

Black box measurement of output gain behavior across spatial coordinates and rendering parameters

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Abstract

This study electrically measured how Sound xR Image maps loudspeaker geometry, Object position, XY mode, Size, Precision, and Distance Attenuation to channel gains, then compared the observations with numerical models. In clearly documented normal output regions, inverse distance weighting with constant power normalization reproduced representative records within approximately 0.005 dB. The five Precision settings were consistent with coefficients corresponding to roughly 8-12 dB of unnormalized attenuation per distance doubling. Speaker ID reassignment, boundary and outside position handling, Object Z, and the reference distance for Distance Attenuation did not reduce to one general rule. The result is a bounded explanatory model for prototyping and further validation, not a claim about the product's undisclosed internal implementation.

Keywords Sound xR Image · spatial audio · black box measurement · DBAP · system identification

1 Introduction

This study followed the deployment of a Sound xR Image system for an outdoor performance. It asks how the renderer converts spatial inputs into measurable output gains and which conditions must be represented to reproduce that behavior.

The objective is to establish repeatable input output relationships and a model with an explicit validity range. It does not attempt to identify or disclose proprietary source code.

2 Methods

A DME7 based testbed was used to vary loudspeaker coordinates and IDs, Object position, XY ON and OFF, Width, Height, Precision, and Distance Attenuation. Output gains were recorded in decibels and retained as condition based blocks in one worksheet.

Measured values were kept separate from candidate equations. Residuals were calculated as measured minus predicted dB, and relative distributions were evaluated after total power normalization where appropriate. The 0.01 dB recording resolution is not an estimate of instrument accuracy or statistical confidence.

3 Candidate Gain Model

Normal output regions were modeled using weights inversely related to the distance D between a calculated rendering position P and each loudspeaker, followed by constant power normalization. The distance exponent a is a candidate coefficient associated with the Precision setting.

The model describes the gain distribution after P has been selected. Whether P equals the input Object position or a projection onto a boundary or internal segment remains a separate unresolved stage.

$$w_i = D_i^{-a}$$

$$g_i = w_i / \sqrt{\sum(w_k^2)}$$

D is the distance from the calculated position to each loudspeaker, and a is the candidate Precision dependent distance exponent.

4 Results

For normal internal positions and the Precision sweep, representative residuals were generally about 0.005 dB. Increasing Precision from Low to High concentrated more gain at the nearest loudspeaker, with coefficients consistent with approximately 8-12 dB of unnormalized attenuation per distance doubling.

In the recent Size experiment, 13 XY ON conditions matched a candidate effective distance expression with a maximum error of approximately 0.0047 dB and an RMS error of approximately 0.0029 dB. Eight limited XY OFF records were explained within approximately 0.0051 dB, but only for Object Z=0.

Changing Speaker ID assignment while preserving the physical coordinate set also changed outputs whose coordinates were not reassigned. One assignment matched the coordinate model within approximately 0.0128 dB, whereas the other retained an error of approximately 1.1728 dB. The behavior therefore cannot be reduced to a simple output column swap.

Several internal and outside positions were explained by projecting the Object onto selected segments before applying the distance model, but the segment selection rule was not established. A recorded Object Z sweep from -2 to +2 showed no gain change, yet the raw block did not identify XY mode and therefore does not verify 3D behavior in XY OFF.

Topic	Primary result	Interpretation range
Normal output	Representative residual about 0.005 dB	Supports distance weighting and power normalization
Precision	About 8-12 dB per distance doubling	Candidate coefficient before normalization
Size	ON max 0.0047 dB OFF max 0.0051 dB	Tested layouts and Object Z=0 only
Speaker ID	Assignment maxima 0.0128 dB and 1.1728 dB	Not explained by coordinates alone
Boundary and Z	Some projections fit Z sweep unchanged	Segment rule and XY OFF remain open

5 Discussion

The measurements strongly support distance based distribution and power normalization within normal operating regions. The effects of Speaker ID and boundary handling nevertheless limit any single DBAP model based only on physical coordinates.

Size sensitivity decreased at higher values, but the data do not support a universal cutoff at 10. Width and Height depended on XY mode, loudspeaker Z, and layout scale. Distance Attenuation appeared as a common offset across valid channels, while its reference distance remains unresolved.

6 Limitations and Further Validation

The present findings are based on static gain records and a limited set of layouts. Channel mapping and state application, an Object Z sweep in XY OFF, asymmetric layouts, boundary transitions, smoothing, and time response require separate measurements.

Future tests should define the Speaker ID rule before measurement and validate it on new layouts. Rotation and scaling of both layout and Object position should then be used to isolate the distance reference used by Distance Attenuation.

7 Conclusion

Normal output regions of Sound xR Image showed close numerical agreement with Precision dependent inverse distance weighting and constant power normalization. Conditional behavior remains for Speaker ID, boundary projection, 3D Object Z, and attenuation distance.

The current result should therefore be used as a bounded prototype model. Extending it to a compatible general model requires independent, pre-registered prediction tests for the unresolved conditions.

Data Availability

The source measurements are provided through the website data preview and the XLSX workbook. Interpretation of the candidate equations is limited to the conditions stated in this report.